BLK II - UNIT 5: METOC Mission Products
e. Watches, Warnings, and Advisories (WWA's)

Introduction to WWA's

The goal of this objective will be to show you have the capability to identify the necessary weather watch, warning, and/or advisory criteria based on provided forecast products. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly apply different thresholds and timelines to forecasted weather events. These foundational concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

Characteristics and Threshold

Criteria

WWA criteria refers to the list of weather events that require WWA input. Some are forecasted events, while others are only issued upon observation. With the goal of WWA criteria being resource protection, the specific criteria may vary between units based on their individual needs and resource thresholds.

Watch

A watch means that hazardous weather is possible. A watch is used when the risk of a hazardous weather event has increased significantly, but its occurrence, location or timing is still uncertain. It is intended to provide enough lead time so those who need to set plans in motion can do so.

Warning

A warning is issued when a hazardous weather event occurs or is imminent. A warning means weather conditions pose a threat to life or property. People in the path of the storm need to take protective action.

Advisory

An advisory is issued when a less serious hazardous weather event is occurring or is imminent. Advisories cause inconvenience and if caution is not exercised, could lead to situations that may threaten life or property.

Timing

The timing of WWA notices are crucial for successful resource and personnel protection. There are several timing thresholds to consider when forecasting and issuing WWAs.

Issue time: This is the time you submit the WWA to external agencies.

Valid times: The forecast period for the WWA criteria to occur with beginning and ending times. Each WWA has its own valid time.

Desired Lead Time (DLT): The minimum required time before the criteria is forecast that you must issue the WWA to alert external agencies to take protective measures.
Lead Time: This is the time between issued time and the first occurrence of the forecast event. You subtract the issue time from the first occurrence time to figure out the lead time.

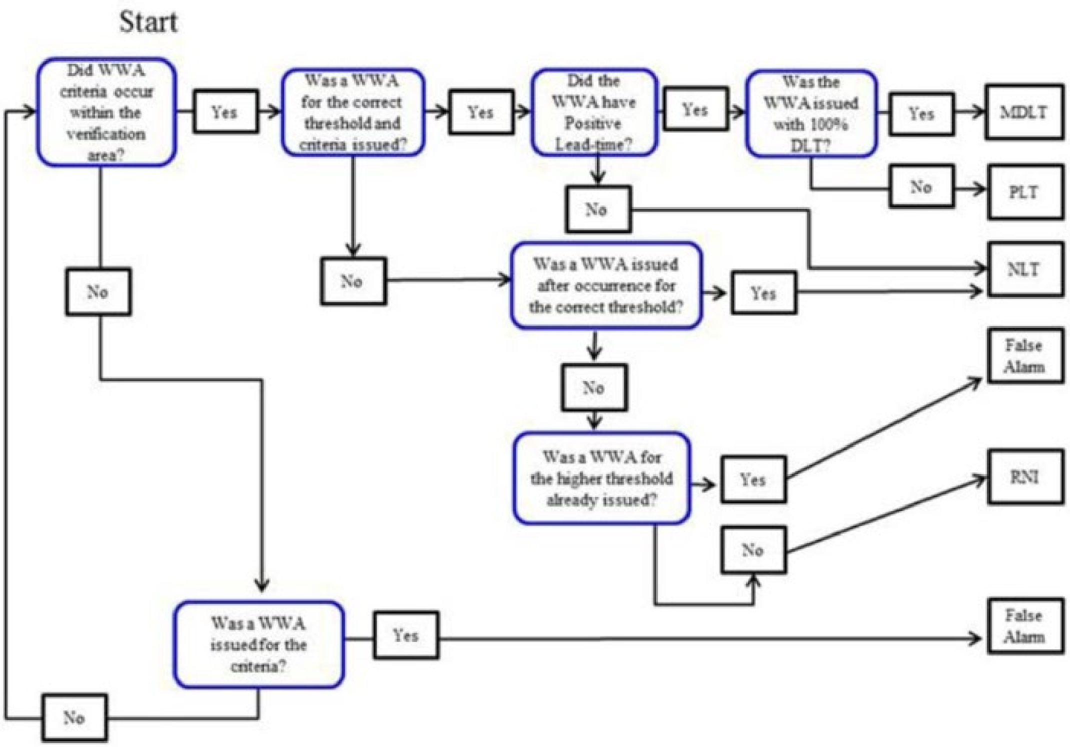
- *Positive Lead Time:* If the first occurrence time is after the issue time, you have a positive lead time.
- Negative Lead Time: If the first occurrence time is before the issue time, you have a negative lead time.

Timing Error: Timing error is the time between the forecast valid time and the actual time of the criterion's first occurrence. Subtract the forecast time from the time of occurrence and if the event occurred later than forecast, then the timing error is positive. If the event occurred before the forecast valid time, then your timing error is negative.

False Alarm: False alarms are when a WWA is issued but the forecasted event never occurred.

Required Not Issued (RNI): An RNI is when there was no WWA criteria forecasted and a WWA was never issued, but one of the WWA criteria occurs. This is the least desirable outcome for forecasters. With a major priority of our job being resource protection, having a significant weather event occur without any agency having warning could result in significant damage to resources or injury to personnel.

WWA Type	Criteria	Desired Lead Time
Warning	Observed Lightning within 5nm	Observed
Warning	Temperature > 90F	6 Hours
Warning	Temperature < 32F	6 Hour
Warning	Non-Convective Winds ≥ 25KT but < 35KT	1 Hour
Warning	Winds ≥ 30KT (Coastal / Ocean Locations Only)	1 Hour
Warning	Moderate Thunderstorms: Hail ≥ ¼" but < ¾" and/or Convective Winds ≥ 35kts but < 50kts	1 Hour
Warning	Severe Thunderstorms: Hail ≥ ¾ in and/or Convective Winds ≥ 50kts	1 Hour
Watch	Lightning within 5nm	30 Min
Watch	Lightning within 10nm	Observed
Watch	Temperature > 90F	12 Hours
Watch	Temperature < 32F	12 Hours
Watch	Non-Convective Winds ≥ 25KT but < 35KT	3 Hours
Watch	Winds ≥ 30KT (Coastal / Ocean Locations Only)	3 Hours
Watch	Moderate Thunderstorms: Hail ≥ ¼" but < ¾" and/or Convective Winds ≥ 35kts but < 50kts	3 Hours
Watch	Severe Thunderstorms: Hail ≥ ¾ in and/or Convective Winds ≥ 50kts	3 Hours
Advisory	Lightning Within 25nm	Observed
Advisory	Temperature ≥ 75F but < 90F	Observed
Advisory	Temperature ≤ 50F but > 32F	Observed
Advisory	Winds ≥ 20KT but <30KT near coastal and ocean areas	Observed



Horizontal Consistency

All issued WWAs must be horizontally consistent with all forecast and observed products we create. These include hourly observations, 5-Day forecasts, Mission Execution Forecasts, Terminal Aerodrome Forecasts, Mission Briefs, and any other product you are required to create.

Failing to maintain horizontal consistency between all your products could result in misinformation while mission planners attempt to adjust for weather considerations. If your products vary in timing or severity of weather events could result in a loss of credibility to your customer. If the products have differing information, this could convey lack of attention to detail that will affect the confidence the customer has in your ability to provide useful or accurate information.

<table>
<tr><th></th><th colspan="4">16-Dec</th><th colspan="4">17-Dec</th><th colspan="4">18-Dec</th><th colspan="4">19-Dec</th><th colspan="4">20-Dec</th></tr>
<tr><td></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr>
<tr><td>Sky Condition</td><td colspan="2">Partly Cloudy</td><td colspan="2">Partly Cloudy</td><td colspan="2">Mostly Cloudy</td><td colspan="2">Overcast</td><td colspan="2">Overcast</td><td colspan="2">Overcast</td><td colspan="2">Overcast</td><td colspan="2">Partly Cloudy</td><td colspan="2">Clear</td><td colspan="2">Clear</td></tr>
<tr><td>Ceiling Height (ft)</td><td colspan="2">N/A</td><td colspan="2">N/A</td><td colspan="2">080</td><td colspan="2">035</td><td colspan="2">020</td><td colspan="2">008</td><td colspan="2">010</td><td colspan="2">N/A</td><td colspan="2">N/A</td><td colspan="2">N/A</td></tr>
<tr><td>Visibility (sm)</td><td colspan="2">10</td><td colspan="2">10</td><td colspan="2">10</td><td colspan="2">10</td><td colspan="2">6</td><td colspan="2">3</td><td colspan="2">3</td><td colspan="2">10</td><td colspan="2">10</td><td colspan="2">10</td></tr>
<tr><td>Temperature (°F)</td><td>High</td><td>64</td><td>Low</td><td>48</td><td>High</td><td>67</td><td>Low</td><td>40</td><td>High</td><td>61</td><td>Low</td><td>40</td><td>High</td><td>43</td><td>Low</td><td>28</td><td>High</td><td>33</td><td>Low</td><td>27</td></tr>
<tr><td>Wind Direction</td><td>↗ 220</td><td>↗ 220</td><td>↗ 220</td><td>↑ 180</td><td>↑ 180</td><td>↑ 170</td><td>↗ 210</td><td>↗ 240</td><td>↗ 240</td><td>↗ 230</td><td>↗ 230</td><td>↗ 240</td><td>↗ 250</td><td>↘ 340</td><td>↓ 350</td><td>↓ 360</td><td>↓ 350</td><td>↓ 360</td><td>↘ 290</td><td>→ 270</td></tr>
<tr><td>Wind Speed (kts)</td><td>5</td><td>5</td><td>5</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>15</td><td>10</td><td>10</td><td>10</td><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td></tr>
<tr><td>Wind Gust (kts)</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>15</td><td>20</td><td>25</td><td>20</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

Mission Execution Forecast

18 Dec 2025	VALID TIME: 0000Z	LOCATION: Keesler AFB / KBIX	FORECASTER: JC

LOCAL AREA FORECAST

TIME	WINDS (KTS)	SKY CONDITION	HAZARDS	VIS	WX	TEMP (F)	DP (F)	ALSTG
0000Z	24010KT	BKN030	LGT ICG 120-250	10 SM	NSW	61	57	29.96
0100Z	24010KT	BKN030	LGT ICG 120-250	10 SM	NSW	60	57	29.95
0200Z	24010KT	BKN030	LGT ICG 120-250	10 SM	NSW	60	58	29.93
0300Z	24010KT	OVC030	LGT ICG 120-250	6 SM	-RA	59	58	29.93
0400Z	24010KT	OVC030	LGT ICG 120-250	6 SM	-RA	58	58	29.93
0500Z	24010KT	OVC030	LGT ICG 120-250	6 SM	-RA	56	55	29.93
0600Z	23010G15KT	OVC030	LGT ICG 120-250	6 SM	-RA	54	52	29.93
0700Z	23010G15KT	OVC025	LGT ICG 120-250	6 SM	-RA	52	51	29.93
0800Z	23010G15KT	OVC025	LGT ICG 120-250	6 SM	-RA	50	49	29.92
0900Z	23010G15KT	OVC025	MOD ICG 100-250	6 SM	-RA	48	47	29.92
1000Z	23010G15KT	OVC020	MOD ICG 100-250	6 SM	-RA	46	45	29.92
1100Z	23010G15KT	OVC020	MOD ICG 100-250	6 SM	-RA	44	43	29.90
1200Z	23010G20KT	OVC020	MOD ICG 100-250	6 SM	-RA	42	41	29.91
1300Z	23010G20KT	OVC020	MOD ICG 100-250	6 SM	-RA	40	38	29.91
1400Z	23010G20KT	OVC020	MOD ICG 100-250	5 SM	-RA	40	39	29.92
1500Z	23010G20KT	OVC020	MOD ICG 100-250	5 SM	-RA	40	39	29.92
1600Z	23010G20KT	OVC015	TSTMS WITHIN 25 SM / MOD ICG	4 SM	-RA	40	39	29.93
1700Z	23010G20KT	OVC015CB	TSTMS WITHIN 10 SM / MOD ICG	4 SM	VCTS RA	41	40	29.94
1800Z	24015G25KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	42	41	29.94
1900Z	24015G25KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	42	42	29.94
2000Z	24015G25KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	42	42	29.94
2100Z	24015G25KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	43	43	29.94
2200Z	24015G25KT	OVC008CB	TSTMS WITHIN 5 SM	3 SM	TSRA	43	43	29.95
2300Z	24015G25KT	OVC008CB	TSTMS WITHIN 5 SM	3 SM	TSRA	43	43	29.95
2400Z	25010G20KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	43	43	29.95
0100Z	25010G20KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	42	42	29.95
0200Z	25010G20KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	41	41	29.96
0300Z	25010G20KT	OVC010CB	TSTMS WITHIN 5 SM	3 SM	TSRA	40	37	29.97
0400Z	25010G20KT	BKN030CB	TSTMS WITHIN 5 SM	3 SM	TSRA	40	35	29.99
0500Z	29010KT	BKN050	TSTMS WITHIN 25 SM	5 SM	-RA	38	31	30.00
0600Z	34010KT	BKN050	NONE	10 SM	NSW	35	25	30.01

TAF KBIX 180000Z 1800/1906 24010KT 9999 BKN030 QNH2993INS 621209 622104
BECMG 1302/1303 23010G15KT 8000 -RA OVC030 QNH2990INS 651209 652104
TEMPO 1316/1318 OVC010CB VCTS
BECMG 1318/1319 24015G25KT 2400 TSRA OVC010CB QNH2994INS
BECMG 1405/1406 33010KT 9999 NSW BKN050 QNH3000INS
TX16/800Z TN04/181700Z